

BIG DATA – Important Exam Questions and Detailed Answers

1. Explain Big Data and its Characteristics (5Vs)

Big Data refers to extremely large datasets that cannot be processed using traditional database systems. The 5Vs are Volume, Velocity, Variety, Veracity, and Value. Volume refers to massive amounts of data, Velocity refers to data generation speed, Variety refers to different data formats, Veracity refers to data quality, and Value refers to useful insights extracted from data.

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2. Explain Hadoop Architecture

Hadoop consists of HDFS, MapReduce, and YARN. HDFS stores data in distributed form, MapReduce processes data in parallel, and YARN manages resources and scheduling. The architecture follows a master-slave model with NameNode and DataNodes.

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3. Explain HDFS Architecture

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4. Explain MapReduce Workflow

MapReduce processes data through Input Splitting, Mapping, Combining, Partitioning, Shuffling, Sorting, Reducing, and Output phases. It enables distributed parallel processing of large datasets.

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5. Explain Hadoop YARN Architecture

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6. Explain Shuffling and Sorting in MapReduce

Shuffling transfers mapper output to reducers. Sorting arranges keys before reducer execution. These phases ensure grouped and organized processing of intermediate data.

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7. Explain Word Count MapReduce Program

Word Count is a basic MapReduce application. Mapper emits pairs and Reducer aggregates counts for each word. It demonstrates the complete MapReduce lifecycle.

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8. Explain Data Mining and Big Data

Data mining extracts patterns and knowledge from data. Big Data technologies provide scalable infrastructure for mining large and complex datasets efficiently.

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9. Explain Advantages and Challenges of Big Data

Advantages include better decision-making, scalability, cost reduction, and predictive analytics. Challenges include security, privacy, storage complexity, and processing overhead.

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10. Explain Hadoop Ecosystem

The Hadoop ecosystem includes HDFS, MapReduce, YARN, Hive, Pig, HBase, Sqoop, Flume, Oozie, and ZooKeeper. These tools support storage, processing, querying, and data movement.

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Important 13-Mark Topics for Exams

- Big Data Characteristics (5Vs)
- Hadoop Architecture
- HDFS Architecture and Working
- MapReduce Workflow
- Shuffling and Sorting
- Word Count Program
- YARN Architecture
- Hive Architecture
- Pig Architecture
- HBase Architecture
- Big Data Applications
- Advantages and Challenges of Big Data